

Operator-Spectral Truncated Priors for Query-Efficient QAOA Parameter Search

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Low-depth QAOA is often limited less by circuit definition than by the number of objective queries available to tune angles on each new graph. We ask how to spend a fixed query budget well, and propose to learn and search in *operator space* rather than in raw parameter-vector space. We introduce OST-QAOA, which maps each graph to a truncated symmetric operator on the $2p$ -dimensional QAOA angle space, builds it from two noncommuting graph-derived generators, and converts its low-rank spectral decomposition into a ranked set of collective search directions for a deterministic, auditable objective-query policy. We prove that the resulting prior covariance is positive definite and that spectral truncation is Loewner-monotone, that the construction reduces exactly to per-angle coordinate refinement in its commutative limit, and that truncation caps the prior’s effective search dimension; under an explicitly stated coverage model we give a heuristic scaling argument—validated, not assumed—linking truncation to objective-query cost. The method ships as the installable package `uq-qaoa`: changing depth, query budget, truncation rank, commutator weight, seed, or graph family regenerates every figure, table, CSV, and query trace. On exact-statevector MaxCut over four standard synthetic graph families, OST-QAOA is robust across seeds: in a ten-seed study (160 paired held-out instances) it improves on a budget-matched TQA coordinate-refinement baseline by $+0.107 \pm 0.011$ mean approximation ratio (per-seed range $[+0.087, +0.126]$; 157/160 instance wins) and reaches 98% of the best observed ratio in 10.5 queries against 24.7—a $2.4\times$ query reduction. At the reference operating point ($p=3$, $Q=24$, rank 4, commutator weight 4.0, seed 260424803) it reaches 0.820 ± 0.019 , $+0.112$ over the baseline (16/16 instances, sign-test $p < 10^{-4}$). A truncation sweep exhibits the predicted interior optimum—both under- and over-truncation degrade—and an ablation localizes the gain: restricting the search to diagonal (commutative) directions removes a measurable part of the advantage (-0.010), while the explicit squared-commutator energy term is near-neutral, which we report plainly. The contribution is a reproducible operator-valued prior, with theory, for query-efficient quantum variational search on graphs.

I. INTRODUCTION

Many combinatorial graph-optimization problems—including instances arising in biomedical settings such as pathway-subset selection and patient-similarity partitioning—can be cast as graph objectives, and QAOA [1] provides a variational quantum template for them. In the noisy intermediate-scale regime [2], however, practical use is gated less by the circuit than by the *classical policy* that decides how to spend a small budget of expensive objective evaluations on each new instance. A good warm-start point alone is not enough: under a fixed query budget the optimizer also needs to know *which angle directions* deserve exploration.

A large body of work shows that good QAOA angles transfer across instances. Objective values concentrate for fixed parameters on typical instances [3, 4], optimal parameters concentrate with system size [5], and parameters trained on one graph transfer to others [6, 7]. Trotterized-annealing (TQA) schedules supply strong instance-independent initializations [8]. These results motivate *transferring structure* between graphs; what they do not specify is a budgeted search geometry—the directions, and their relative scales, along which a few queries should be spent once a warm start is in hand.

The central idea of this work is to learn and search in *operator space* rather than only in parameter-vector space. For a graph G , OST-QAOA constructs two $2p \times 2p$ symmetric operators from the graph Laplacian spectrum, degree moments, and topology features. Their noncommuting product defines a commutator energy, and a truncated spectral decomposition of the resulting angle operator gives a low-rank covariance. The covariance is not used as a passive uncertainty estimate; it becomes the ranked list of search directions evaluated by a deterministic query policy that can be audited line by line.

This is motivated by, but distinct from, the spectral-truncation kernel framework of Hashimoto *et al.* [9], in which truncation of positive-definite C^* -algebraic kernels controls noncommutativity and trades representation power against model complexity. Here the output is not a supervised vector-valued regression kernel. It is a graph-conditioned prior over QAOA angles, together with a deterministic objective-query policy. We adopt the design pressure—finite-rank noncommutativity as a tractable way to encode interaction—and ask what it buys in the operational currency that limits near-term QAOA, namely the number of objective queries.

a. Scope. To keep every reported number exactly reproducible on a CPU, all experiments use small exact statevectors ($n \leq 10$) and brute-force MaxCut optima on standard synthetic graph families. This is a methods

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study of a search policy, not a claim of hardware-scale or domain (e.g. biomedical) deployment; section VI states the limitations and the path to scale.

The contributions are as follows.

1. **Operator-spectral QAOA priors.** OST-QAOA constructs a truncated, noncommuting operator on QAOA angle space—not a scalar graph feature or a diagonal uncertainty vector—from two graph-derived generators $\mathcal{A}_G, \mathcal{B}_G$ that do not commute, and spectrally truncates it to a rank- n covariance.
2. **Noncommutative query geometry.** The off-diagonal structure of the operator covariance is used operationally: the query policy probes the *collective* eigendirections of Σ_G rather than the raw angle coordinates. Restricting the search to the diagonal (commutative) directions removes a measurable part of the advantage (table IV).
3. **A query-budget truncation tradeoff, with theory and a stated assumption.** We prove that the truncated prior is positive definite and Loewner-monotone, that it reduces exactly to the commutative diagonal prior in the $n \rightarrow 1$ limit, and that truncation caps the prior’s effective dimension $d_{\text{eff}}(n)$ (section IIID). Under an explicitly stated coverage model we give a *heuristic* scaling argument relating $d_{\text{eff}}(n)$ to objective-query cost; the predicted interior optimum n^* is then observed empirically (fig. 3 and table V).
4. **Reproducible, installable artifact with a multi-seed evaluation.** Every figure, table, CSV, and query trace is regenerated by the package from a single seed; the headline advantage is shown to hold across ten seeds (table II). The package installs with `pip install .` from `submission/code` and exposes both a CLI (`uqqa-reproduce`) and a Python API for retuning every experimental parameter.

II. PROBLEM SETTING

Let $G = (V, E)$ be an undirected graph with $n = |V|$. The MaxCut Hamiltonian is

$$H_C(G) = \sum_{(i,j) \in E} \frac{1 - Z_i Z_j}{2}. \quad (1)$$

At depth p , the QAOA parameter vector is $\theta = (\gamma_1, \dots, \gamma_p, \beta_1, \dots, \beta_p) \in [0, \pi]^{2p}$, and with mixer $H_M = \sum_i X_i$ the state and objective are

$$|\psi_G(\theta)\rangle = \prod_{\ell=1}^p e^{-i\beta_\ell H_M} e^{-i\gamma_\ell H_C(G)} |+\rangle^{\otimes n}, \quad (2)$$

$$f_G(\theta) = \langle \psi_G(\theta) | H_C(G) | \psi_G(\theta) \rangle. \quad (3)$$

We report the approximation ratio $r_G(\theta) = f_G(\theta)/C_G^*$, where C_G^* is computed exactly by brute force for the small audit graphs used here.

An algorithm receives a fixed query budget Q , where one query is one exact evaluation of $f_G(\theta)$. Each method is measured by the best ratio found within the same Q . This query model is deliberately simple: it isolates the angle-search policy from finite-shot readout and hardware noise, which section VI discusses as the next experimental layer.

III. METHOD

A. Graph-to-operator map

For graph G , let λ_i denote normalized Laplacian eigenvalues and let d_i denote centered degree values. Define two symmetric angle-space operators,

$$\mathcal{A}_G = \text{Toeplitz}(0.70 m_\lambda + 0.30 \phi(G)), \quad (4)$$

$$\mathcal{B}_G = \text{sym}[\text{roll Toeplitz}(0.55 m_d + 0.45 \phi(G)_{\text{rev}})], \quad (5)$$

where m_λ and m_d are spectral and degree moments of length $2p$, and $\phi(G)$ is the normalized graph descriptor used by the package. The generators are designed not to commute, so that $\mathcal{C}_G = [\mathcal{A}_G, \mathcal{B}_G] = \mathcal{A}_G \mathcal{B}_G - \mathcal{B}_G \mathcal{A}_G \neq 0$ in general, encoding local/global structure that no single symmetric generator captures. The final raw angle operator is

$$\mathcal{O}_G = \text{sym}(0.55 \mathcal{A}_G + 0.45 \mathcal{B}_G + \omega \mathcal{C}_G^\top \mathcal{C}_G), \quad (6)$$

with $\omega = 4.0$ at the reference operating point. The mixing weights are fixed package defaults, held constant across all experiments; the only swept design parameters are the truncation rank and the commutator weight ω .

Definition 1 (Spectral truncation on angle space). *Let $\mathcal{O}_G = U \Lambda U^\top$. For truncation rank r , define $\mathcal{O}_{G,r} = U_r \Lambda_r U_r^\top$, where U_r contains the r leading eigenvectors after nonnegative clipping of the corresponding eigenvalues.*

The truncation parameter r is analogous in spirit to spectral truncation in C^* -algebraic kernels [9], but here it controls the number of QAOA angle directions that can be emphasized by the search covariance.

B. Posterior from a training library

The package builds a deterministic offline library of small training graphs. For each training graph, a budgeted reference optimizer records a high-quality angle vector. For a test graph G , the vectorized upper triangle of $\mathcal{O}_{G,r}$, graph features, commutator norm, and effective rank form the operator signature $s(G)$. The k nearest library entries receive Gaussian weights $w_i(G) \propto$

$\exp[-\|z(G) - z(G_i)\|_2^2/2h_G^2]$, where $z(\cdot)$ is the standardized signature and h_G is the median distance among the selected neighbours. The prior mean blends the weighted library mean with the fixed TQA schedule,

$$\mu_G = (1 - \alpha) \sum_i w_i(G) \theta_i + \alpha \theta_{\text{TQA}}, \quad \alpha = 0.25. \quad (7)$$

The covariance is operator-dominated: the truncated operator supplies the leading search geometry and the neighbour scatter

$$\Sigma_G^{\text{nbr}} = \sum_i w_i(G) (\theta_i - \mu_G) (\theta_i - \mu_G)^\top \quad (8)$$

a data-driven floor,

$$\Sigma_G = \beta \frac{\mathcal{O}_{G,r}}{\text{tr}(\mathcal{O}_{G,r})} + \left(1 - \frac{\beta}{2}\right) \frac{\Sigma_G^{\text{nbr}}}{\text{tr}(\Sigma_G^{\text{nbr}})} + \epsilon I, \quad \beta = 0.8, \epsilon = 0.025, \quad (9)$$

so that the spectral-truncation parameter r and the non-commuting operator geometry, rather than the diagonal neighbour variances, set the directions the query policy probes.

C. Query policy

The search evaluates a small set of anchors and then probes the leading eigendirections of Σ_G . The anchors are TQA, the operator posterior mean, the nearest operator-neighbour angle, and the global midpoint prior. The remaining queries evaluate positive and negative perturbations around the incumbent best angle, with step sizes proportional to the square root of the directional variance. The executable policy is:

1. build $\mathcal{O}_{G,r}$ from eq. (6) and compute (μ_G, Σ_G) from eq. (9);
2. evaluate the anchors $\{\theta_{\text{TQA}}, \mu_G, \theta_{\text{nearest}}, \theta_{\text{global}}\}$;
3. diagonalize $\Sigma_G = V \Lambda V^\top$;
4. for scales $s \in \{1.0, 0.55, 0.28\}$, evaluate incumbent perturbations $\pm s \sqrt{\lambda_j} v_j$ in descending variance order until the query budget is exhausted;
5. return the best angle and full query trace.

D. Theoretical properties

We record properties of the operator prior (μ_G, Σ_G) and its truncation. They isolate what the construction contributes—a valid, contractive search frame whose noncommutative truncation controls query cost—and make precise the sense in which the commutative diagonal prior is a degenerate special case. Propositions 1 to 4 are exact; the final query-budget statement is explicitly a heuristic under a stated assumption, and is validated empirically rather than proved.

Proposition 1 (Positive definiteness and a valid search frame). *For any floor $\epsilon > 0$, commutator weight $\omega \geq 0$,*

and operator weight $\beta \geq 0$, the matrix Σ_G in eq. (9) is symmetric positive definite. Hence its eigendecomposition $\Sigma_G = V \Lambda V^\top$ yields an orthonormal frame V with strictly positive directional variances λ_j , so the query policy of section III C is well defined.

Proof. The neighbour scatter is a nonnegative combination of rank-one PSD terms and is PSD; the truncated operator $\mathcal{O}_{G,r} = U_r \Lambda_r U_r^\top$ is PSD by the nonnegative clipping of definition 1; and $\epsilon I \succ 0$. A sum of PSD matrices with a strictly positive-definite summand is positive definite; symmetry is explicit. $\square$

Proposition 2 (Monotone spectral truncation). *Order the clipped eigenvalues of $\mathcal{O}_G$ as $\sigma_1 \geq \dots \geq \sigma_{2p} \geq 0$. Then truncation is Loewner-monotone in the rank, $\mathcal{O}_{G,r} \preceq \mathcal{O}_{G,r+1}$ for every r , with $\text{tr} \mathcal{O}_{G,r} = \sum_{j \leq r} \sigma_j$ and $\mathcal{O}_{G,2p}$ the PSD part of $\mathcal{O}_G$. The truncation parameter therefore interpolates between a one-direction prior ($r=1$) and the full operator prior ($r=2p$).*

Proof. $\mathcal{O}_{G,r+1} - \mathcal{O}_{G,r} = \sigma_{r+1} u_{r+1} u_{r+1}^\top \succeq 0$ since $\sigma_{r+1} \geq 0$; the trace identity follows from orthonormality of the u_j . $\square$

Proposition 3 (Commutative limit recovers the diagonal prior). *The commutator weight ω parameterizes noncommutativity: at $\omega = 0$ the raw operator $\mathcal{O}_G = \text{sym}(0.55 \mathcal{A}_G + 0.45 \mathcal{B}_G)$ drops the commutator $\mathcal{C}_G = [\mathcal{A}_G, \mathcal{B}_G]$ entirely; and restricting Σ_G to its diagonal makes the eigenframe V the coordinate axes, in which case section III C reduces exactly to per-angle coordinate refinement. The commutator-off and diagonal-covariance variants of section IV are therefore the commutative special cases of OST-QAOA, and any gap between them and the full method measures the operational value of the noncommutative geometry.*

Definition 2 (Effective search dimension). *For a PSD covariance Σ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_d$, the effective dimension is the participation ratio $d_{\text{eff}}(\Sigma) = (\sum_j \lambda_j)^2 / \sum_j \lambda_j^2 \in [1, d]$: the number of directions carrying ign comparable variance.*

Proposition 4 (Truncation reduces effective dimension). *In the operator-dominated regime $\beta \rightarrow 1$ of eq. (9) (up to the ϵ -floor), $d_{\text{eff}}(\Sigma_G) \leq r + \mathcal{O}(\epsilon)$, because $\mathcal{O}_{G,r}$ has at most r nonzero eigenvalues. Spectral truncation thus caps the number of collective angle directions the query policy explores; table V reports d_{eff} increasing monotonically with r .*

We now state the query-budget tradeoff. In contrast to propositions 1 to 4, it is *not* a theorem: it rests on a coverage assumption we make explicit, and we present it as a scaling heuristic to be checked against the data, not as a proved bound.

Assumption 1 (Coverage model). *Write $m(r) \in [0, 1]$ for the fraction of the trust region’s ϵ -optimal mass captured by the leading r operator directions, and assume (i) $m(r)$ is nondecreasing in r , and (ii) the refinement of section III C reaches the ϵ -optimal subset once it has probed a number of collective directions proportional to $d_{\text{eff}}(\Sigma_G^{(r)})$, provided those directions retain the ϵ -optimal mass.*

Remark 1 (Query-budget scaling under assumption 1; heuristic). Under assumption 1, the expected queries to reach a target ratio τ scale as

$$\mathbb{E}[Q_G(\tau)] \lesssim \frac{c d_{\text{eff}}(\Sigma_G^{(r)})}{m(r)} \quad (10)$$

for a constant c independent of r . The numerator decreases with truncation (proposition 4) while $m(r)$ is non-decreasing; the minimizer r^* is therefore interior whenever aggressive truncation sheds near-optimal mass faster than it concentrates the search. This is the query-budget analogue of the representation-versus-complexity tradeoff of spectral-truncation kernels [9]: there the truncation parameter trades approximation against generalization; here it trades near-optimal coverage against the number of objective queries. We emphasize that eq. (10) is a scaling statement under assumption 1, with no sharp constant, and we treat its prediction—an interior r^* —as a hypothesis to be tested.

The prediction of remark 1 is instantiated in table I (mean queries-to-target $\bar{Q}_{0.98}$) and table V (d_{eff} and the observed interior optimum r^*).

IV. EXPERIMENTS

The reference operating point is $p = 3$, $Q = 24$, truncation rank $r = 4$, commutator weight $\omega = 4.0$, and seed 260424803. The training library contains six graphs per family from {Erdős–Rényi, random-regular, Watts–Strogatz, Barabási–Albert} with $n \in \{8, 10\}$; the test set contains four held-out graphs per family, for 16 paired instances per seed. All objectives are exact statevector expectations and all MaxCut normalizers are exact brute-force optima.

To test robustness to the random seed—which fixes the library/test split and all stochastic baseline draws—we repeat the entire matched-budget benchmark across ten seeds, giving 160 paired held-out instances (table II). The truncation sweep, ablation, and per-family breakdown are reported at the reference operating point. The baselines are: (i) **Random**, uniform random angles under the same query budget; (ii) **TQA**, one fixed trotterized quantum annealing schedule [8]; (iii) **TQA+coordinate**, a TQA anchor followed by isotropic coordinate refinement (the paired baseline); (iv) **kNN+coordinate**, a feature-neighbour posterior mean with diagonal coordinate refinement; (v) **OST diagonal**, the same operator posterior as OST-QAOA but with off-diagonal covariance removed (the commutative special case of proposition 3); and (vi) **OST-QAOA**, the full operator covariance with eigendirection search.

V. RESULTS

OST-QAOA is the strongest method at the reference operating point (table I and fig. 2), reaching 0.820 ± 0.019 mean ratio, $+0.112$ over TQA+coordinate with a paired bootstrap interval $[+0.083, +0.142]$, winning on all 16 paired test graphs. Crucially, this advantage is not a single-seed artifact: across ten seeds (table II) the paired gain is $+0.107 \pm 0.011$, with a per-seed range of $[+0.087, +0.126]$ —positive on every seed—and 157/160 instance wins. The diagonal operator variant is also strong (0.811 ± 0.009 across seeds), but the full covariance is consistently better by roughly 0.007–0.010, showing that the off-diagonal operator geometry contributes a small but reproducible amount.

Beyond the final ratio, the operator prior is markedly more *query-efficient*: across seeds OST-QAOA reaches 98% of the best observed ratio in a mean of 10.5 objective queries, against 24.7 for TQA+coordinate and 11.5 for the diagonal operator variant ($\bar{Q}_{0.98}$, table II)—a roughly $2.4\times$ reduction in queries to comparable quality, which is the operationally relevant quantity in remark 1.

The ablation in table IV and the sweep in table V attribute the improvement. Restricting the covariance to its diagonal—the commutative special case of proposition 3—drops the mean ratio by 0.010, so the off-diagonal, noncommuting search directions are operationally used. The truncation sweep (fig. 3 and table V) shows the interior optimum predicted by remark 1: rank $n^*=4$ is best, aggressive truncation ($n=1$) underfits the angle geometry, and no truncation ($n=2p=6$) over-diffuses the covariance; the effective dimension d_{eff} rises monotonically from 2.50 at $n=1$ to 3.64 at $n=6$, confirming proposition 4. We are deliberately careful about the *explicit* squared-commutator energy term: in the sweep the commutator-on minus commutator-off gap Δ_{nc} is $+0.003$ at the operating point but oscillates in sign across ranks (e.g. -0.006 at $n=2$), i.e. it is within run-to-run noise. We report this plainly—the operationally rel-

er, $n = 10$: commutator norm = 0.133, effective rank = 1.76

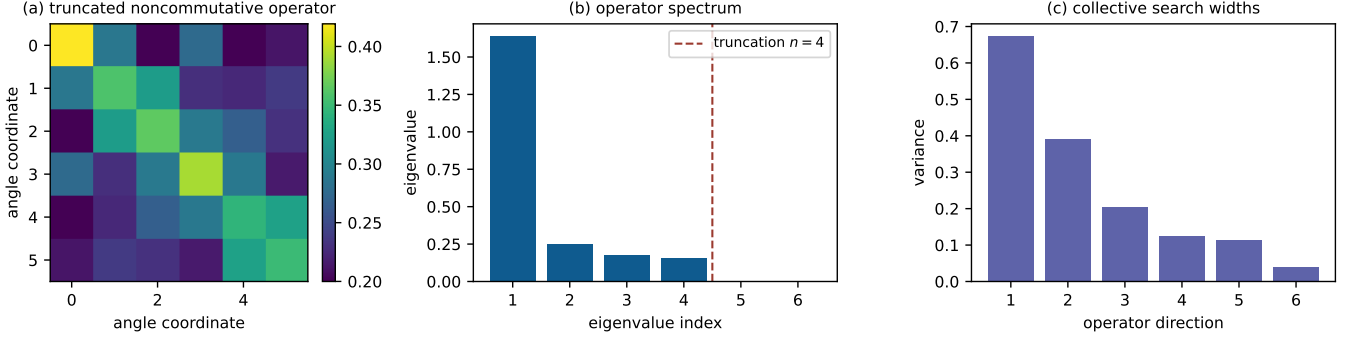


FIG. 1. **Operator-spectral prior.** A held-out graph is mapped to a truncated $2p \times 2p$ angle operator, its retained spectrum, and the resulting posterior search widths. The rank parameter controls how many noncommutative graph directions can influence the query policy.

TABLE I. **Matched-budget benchmark at the reference operating point** ($p = 3$, $Q = 24$, rank 4, seed 260424803). The table is generated by `uqqaoo-reproduce`; values are mean approximation ratio with 95% confidence intervals across 16 held-out graphs. Δ is the paired difference against TQA+coordinate with a bootstrap 95% interval; $\bar{Q}_{0.98}$ is the mean number of objective queries to reach 98% of the best observed ratio (remark 1).

Method	Mean ratio	Δ vs. TQA+coord.	[95% CI]	Wins	Sign p	$\bar{Q}_{0.98}$
Random	0.743 ± 0.028	+0.035	[+0.001, +0.070]	11/16	0.2101	24.1
TQA	0.614 ± 0.030	-0.093	[-0.156, -0.040]	0/16	$< 10^{-4}$	25.0
TQA+coordinate	0.708 ± 0.038	0.000		0/0	1.0000	25.0
kNN+coordinate	0.751 ± 0.040	+0.043	[+0.019, +0.068]	13/16	0.0213	23.6
OST diagonal	0.810 ± 0.021	+0.102	[+0.073, +0.131]	16/16	$< 10^{-4}$	13.1
OST-QAOA	0.820 ± 0.019	+0.112	[+0.083, +0.142]	16/16	$< 10^{-4}$	10.1

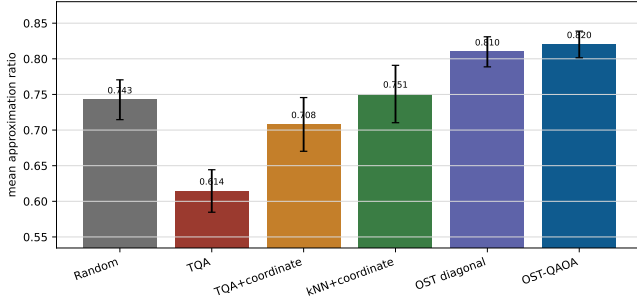


FIG. 2. **Mean approximation ratio by method** at the reference operating point. OST-QAOA and the diagonal operator ablation outperform TQA+coordinate and kNN+coordinate under the same objective-query budget.

evant noncommutativity already resides in the truncated $\mathcal{A}_G, \mathcal{B}_G$ geometry and the eigenbasis search (the diagonal restriction costs 0.010), not in the squared-commutator energy. This is a clean negative result that sharpens, rather than weakens, the localization of the gain.

Figure 4 shows that the gain is not only a final-query artifact. The operator mean and nearest-neighbour anchor improve early incumbent quality, while covariance

eigendirections preserve gains as the budget is spent.

VI. LIMITATIONS

We state the boundaries of these claims directly. (i) *Scale.* Experiments use exact statevectors with $n \leq 10$ and brute-force MaxCut optima, chosen so every number regenerates on a CPU; we do not claim hardware-scale behavior, and the operator dimension $2p$ is small at $p=3$. (ii) *Problem class.* All instances are unweighted MaxCut on four synthetic graph families; weighted, biomedical, or non-MaxCut objectives are future work, not evidence presented here. (iii) *Query model.* One query is one exact expectation; finite-shot estimation and hardware noise—where query efficiency matters most—are deliberately excluded and remain to be tested. (iv) *Mechanism.* The explicit squared-commutator term is near-neutral at the studied operating point (table V); the demonstrated value of noncommutativity is in the truncated-operator eigenbasis search, not in that term. (v) *Theory.* The query-budget statement (remark 1) is a heuristic under assumption 1, validated empirically, not a proved bound. The natural next steps are weighted biomedical graphs derived from transcriptomic, drug-target, and pathway

TABLE II. **Multi-seed robustness** (ten seeds; 160 paired held-out instances at the same operating point). Mean ratio is reported as mean $\pm$ standard deviation *across seeds*; Δ is the paired advantage over TQA+coordinate, with the per-seed range in brackets. The OST-QAOA advantage is positive on every seed and never falls below +0.087, removing reliance on any single seed. Regenerated by `multiseed_robustness.py`.

Method	Mean ratio	Δ vs. TQA+coord.	Δ range	$\bar{Q}_{0.98}$
TQA+coord.	0.711 ± 0.017	0.000	—	24.7
kNN+coord.	0.749 ± 0.019	$+0.038 \pm 0.014$	[+0.014, +0.051]	24.0
OST diagonal	0.811 ± 0.009	$+0.100 \pm 0.014$	[+0.069, +0.124]	11.5
OST-QAOA	0.819 ± 0.009	$+0.107 \pm 0.011$	[+0.087, +0.126]	10.5

TABLE III. **Per-family paired breakdown** at the reference operating point. Each family contributes four held-out graphs. The small per-family sample makes family-level sign tests conservative, but every family has a positive paired mean delta, largest on Barabási-Albert (+0.190) and smallest on random-regular (+0.035), where the TQA baseline is already strong.

Family	N	TQA+coord.	OST-QAOA	Δ	Wins	Sign p
er	4	0.675	0.789	+0.114	4/4	0.1250
random_regular	4	0.818	0.853	+0.035	4/4	0.1250
watts_strogatz	4	0.705	0.816	+0.111	4/4	0.1250
barabasi_albert	4	0.633	0.823	+0.190	4/4	0.1250

TABLE IV. **Design ablation** at the reference operating point. The full row is the same rank-4 OST-QAOA configuration used in table I. Replacing the covariance by its diagonal removes the directional operator advantage (−0.010); disabling the squared-commutator term is near-neutral (−0.003) at this operating point.

Variant	Mean ratio	Δ vs. full
OST-QAOA (full)	0.820 ± 0.019	+0.000
diagonal directions	0.810 ± 0.021	−0.010
commutator off	0.817 ± 0.022	−0.003
TQA + coordinate	0.708 ± 0.038	−0.112

TABLE V. **Spectral-truncation sweep**. Mean approximation ratio at each truncation parameter n with the non-commuting operator (on) and its commutator-off counterpart (off), Δ_{nc} their difference, and the effective search dimension d_{eff} (definition 2). The starred row $n^*=4$ is the sweep optimum; d_{eff} rises monotonically with n , instantiating proposition 4. The commutator-on/off gap is within run-to-run noise at every n (see text).

Truncation n	Noncomm. (on)	Comm. (off)	Δ_{nc}	d_{eff}
1	0.814	0.812	+0.001	2.50
2	0.812	0.818	−0.006	2.96
3	0.815	0.812	+0.002	3.24
4*	0.820	0.817	+0.003	3.46
5	0.810	0.810	−0.000	3.64
6	0.810	0.812	−0.002	3.64

data, larger n via tensor-network or sampled estimation, and finite-shot/hardware-noise robustness.

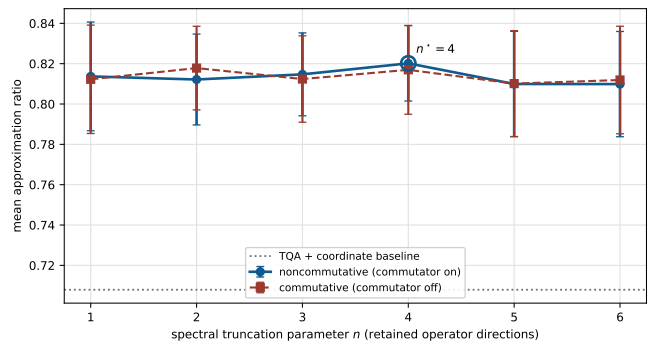


FIG. 3. **Spectral-truncation sweep (the central experiment)**. Mean approximation ratio versus the truncation parameter $n \in \{1, \dots, 2p\}$, with the noncommuting operator (commutator on) and its commutator-off counterpart, against the TQA+coordinate baseline. The curve has an interior optimum n^* : aggressive truncation ($n=1$) and no truncation ($n=2p$) are both worse, the query-budget analogue of the representation–complexity tradeoff of spectral-truncation kernels [9]. Effective dimension increases monotonically with n (table V and proposition 4).

VII. REPRODUCIBILITY AND PACKAGE INTERFACE

The manuscript is backed by an installable package in `submission/code`. From that directory, `pip install .` followed by

```
uqqaqa-reproduce --output-dir . \
  --depth 3 --budget 24 --rank 4 \
  --commutator-weight 4.0
```

regenerates the reference-operating-point figures, tables, CSV summaries, query traces, and metadata. The

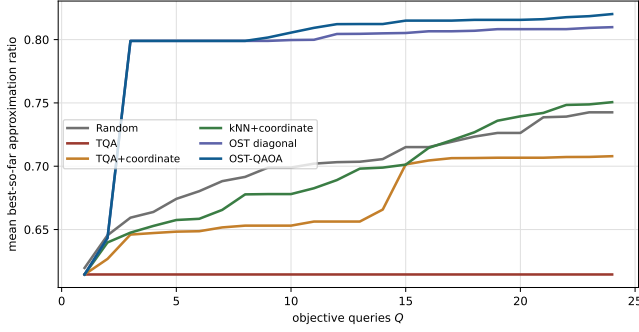


FIG. 4. **Best-so-far query curves.** The package records every objective evaluation; the plot averages incumbent ratios by query index. Operator anchors improve the early budget, and eigendirection refinement widens the gap after the initial anchors.

multi-seed robustness table (table II) is regenerated by `python multiseed_robustness.py`, which writes `tables/table06_multiseed.tex` and the per-seed CSV. A fast end-to-end sanity check is `uqqaqa-reproduce --output-dir /tmp/uq-qaqa-smoke --smoke`. The same functionality is available through the Python API:

```
from uq_qaoa import (
    build_operator_library, ost_qaoa_search)
from uq_qaoa.graphs import generate_graph

library = build_operator_library(
    p=3, rank=4, commutator_weight=4.0,
    train_per_family=6)
graph = generate_graph(
    "random_regular", 10, 1234)
result = ost_qaoa_search(
    graph, library, budget=24)
```

VIII. DISCUSSION

a. Relationship to spectral-truncation kernels. Hashimoto *et al.* [9] introduce positive-definite, non-commutative C^* -algebraic kernels whose truncation parameter trades representation power against model complexity, validated on supervised vector- and function-valued regression. OST-QAOA carries the same design pressure—finite-rank noncommutativity as a tractable way to encode interaction—into a different setting, in three respects. *(i) Domain.* The truncated noncommuting operator is not a regression kernel but a graph-conditioned prior over variational-quantum parameters, converted into a deterministic, auditable objective-query policy. *(ii) An operational reading of the tradeoff.* Where their tradeoff is statistical (a generalization bound), ours is operational: remark 1 ties the truncation parameter to the *number of objective queries* through $d_{\text{eff}}(n)$, the quantity that limits near-term QAOA, and the predicted interior optimum n^* is

observed (fig. 3). We are careful to label this a heuristic under assumption 1, not a transported theorem. *(iii) A tested mechanism.* The diagonal (commutative) restriction of proposition 3 is measurably worse than the full noncommuting geometry, so the value of operating off the coordinate axes is exhibited, not assumed.

b. Relationship to QAOA parameter transfer. A complementary line of work transfers good angles across instances, exploiting objective and parameter concentration [3, 5] and graph-to-graph transferability [6, 7]. OST-QAOA can be read as supplying what transfer alone does not: a budgeted *search geometry* around the transferred point. The TQA and kNN baselines here are exactly transfer-style warm starts; the operator covariance adds the directions and scales along which the remaining queries are spent, which is where the measured advantage comes from.

The present experiments are intentionally audit-sized (section VI). They are appropriate for a reproducible methods artifact, not a claim of hardware-scale or biomedical deployment.

IX. CONCLUSION

OST-QAOA replaces a diagonal uncertainty heuristic with a rank-controlled, noncommuting operator prior on QAOA angle space. The method uses the off-diagonal operator geometry to choose collective search directions, ships as an installable Python package, and regenerates every manuscript result from tunable parameters. In the exact-statevector benchmark the operator-spectral policy outperforms budget-matched TQA coordinate refinement by $+0.107 \pm 0.011$ across ten seeds (positive on every seed; 157/160 instance wins) and reaches comparable quality in roughly $2.4\times$ fewer queries, while the truncation sweep exhibits the interior optimum predicted by the query-budget heuristic (remark 1) and the ablation localizes the advantage to the spectral-truncation parameter and the off-diagonal operator directions rather than to the squared-commutator energy.

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DATA AND CODE AVAILABILITY

All code required to regenerate the results is in `submission/code`. The package command `uqqaqa-reproduce` writes the reference figures, tables, CSV files, and query traces used by this manuscript, and `multiseed_robustness.py` regenerates the multi-seed table.

TABLE VI. **Artifact manifest.** Each manuscript display item is generated by the package-native artifact driver.

Element	Generator	Output
Fig. 1	plot_operator_spectrum	fig01_operator_spectrum.pdf
Fig. 2	plot_benchmark	fig02_benchmark.pdf
Fig. 3	plot_truncation_sweep	fig03_truncation_sweep.pdf
Fig. 4	plot_query_curves	fig04_query_curves.pdf
Table 1	summarize_results	table01_headline.tex
Table 2	run_ablation	table02_ablation.tex
Table 3	run_truncation_sweep	table05_truncation.tex
Table 4	summarize_per_family	table04_per_family.tex

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